

Q1.

### Question 1

How to Attempt?

Write an SQL query to display:

The transaction id, transaction amount and transaction type of all the transactions whose transaction type is "Debit" and transaction amount is greater than 10000 but less than 50000

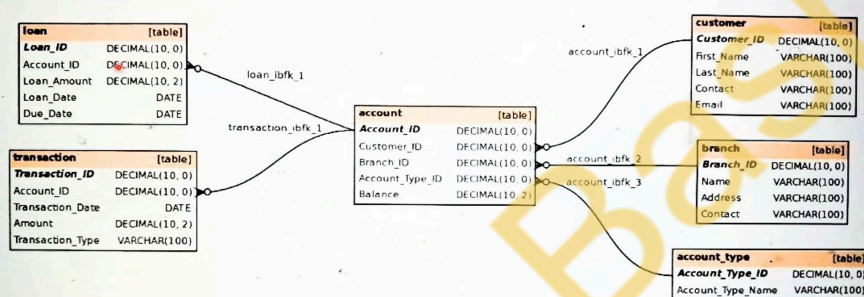
Your output should have 3 columns as given below:

| TRANSACTION_ID | AMOUNT | TRANSACTION_TYPE |
|----------------|--------|------------------|
|----------------|--------|------------------|

You can view the database schema by clicking the View Schema tab at the bottom of the question window on the right-hand side of the screen.

Enter your Response

View Schema



Answer :

```
SQL Query:
sql
SELECT
    Transaction_ID,
    Amount,
    Transaction_Type
FROM
    transaction
WHERE
    Transaction_Type = 'Debit'
    AND Amount > 10000
    AND Amount < 50000;
```

Q2.

## Question 2

### How to Attempt?

Write an SQL query to display:

The first name, contact number and balance of all the customers whose account type starts with "Sa". The output should be ordered by the customer's first name.

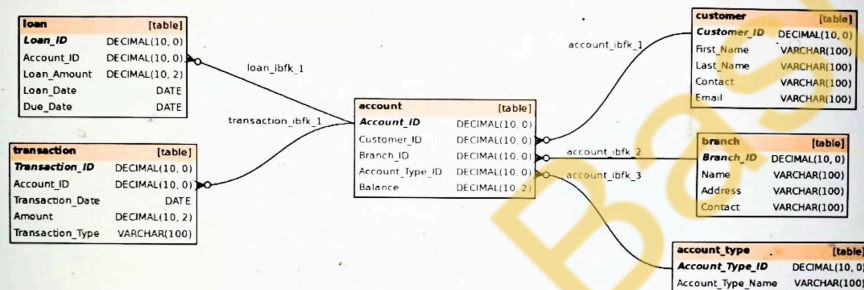
Your output should have 2 columns as given below:

| FIRST_NAME | CONTACT | BALANCE |
|------------|---------|---------|
|------------|---------|---------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

Enter your Response

View Schema



Answer :

```
SQL Query:
sql
SELECT
    c.First_Name,
    c.Contact,
    a.Balance
FROM
    customer c
JOIN
    account a ON c.Customer_ID = a.Customer_ID
JOIN
    account_type at ON a.Account_Type_ID = at.Account_Type_ID
WHERE
    at.Account_Type_Name LIKE 'Sa%'
ORDER BY
    c.First_Name;
```

Q3.

1. Database Query Question

### Question 1

How to Attempt?

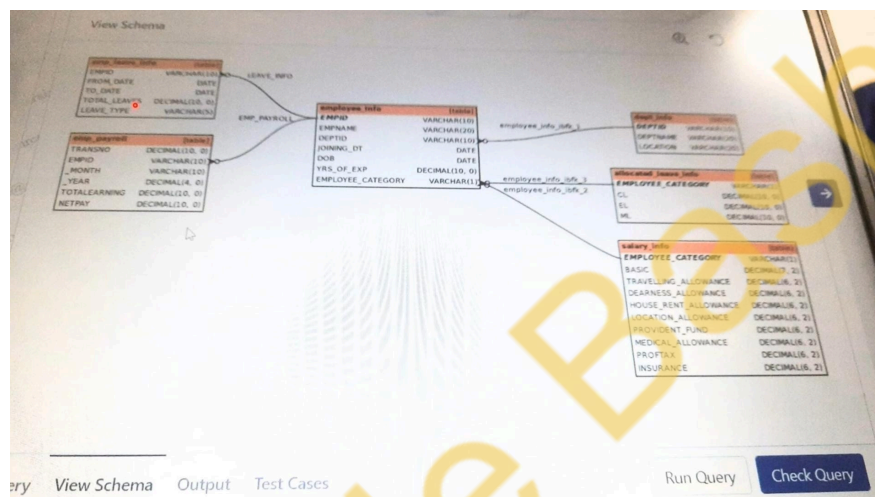
Write an SQL query to display:

The employee ID, Name, Basic salary, and Net pay for employees whose basic salary is greater than 5,000.

Your output should have 4 columns as given below:

| EMPID | EMPNAME | BASIC | NETPAY |
|-------|---------|-------|--------|
|-------|---------|-------|--------|

You can view the database schema by clicking the View Schema tab at the bottom of query window on the right-hand side of the screen.



Answer :

Query:

```
sql Copy code

SELECT
    emp_info.empid AS EMPID,
    emp_info.empname AS EMPNAME,
    salary_info.basic AS BASIC,
    emp_payroll.netpay AS NETPAY
FROM
    emp_info
JOIN
    salary_info ON emp_info.employee_category = salary_info.employee_category
JOIN
    emp_payroll ON emp_info.empid = emp_payroll.empid
WHERE
    salary_info.basic > 5000;
```

Q4.

1. Database Query Question

Question 1

How to Attempt?

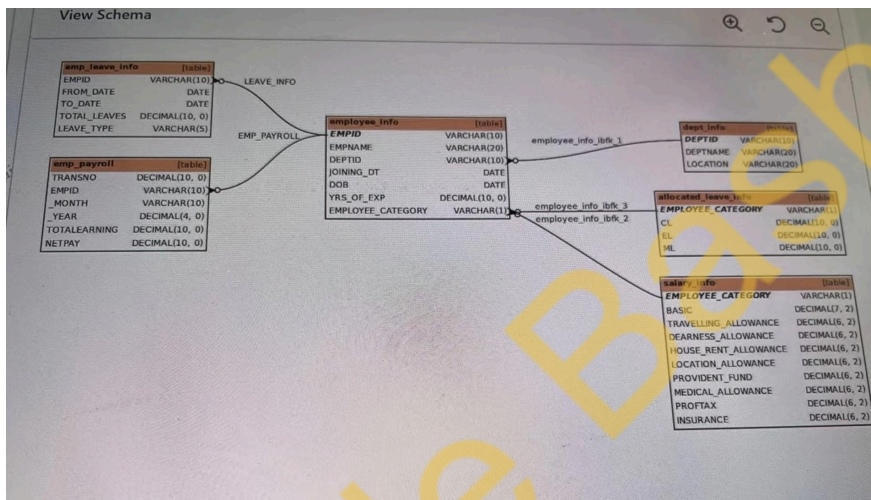
Write an SQL query to display:

The names of employees who have more than 5 years of experience and joined after January 1, 2001. Use alias "Employee ID", "Employee Name"

Your output should have 2 columns as given below:

| Employee ID | Employee Name |
|-------------|---------------|
|-------------|---------------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



Answer :

SQL Query:

```
sql
SELECT
    EMPID AS "Employee ID",
    EMPNAME AS "Employee Name"
FROM
    employee_info
WHERE
    YRS_OF_EXP > 5
    AND JOINING_DT > '2001-01-01';
```

Q5.

1. Database Query Question

### Question 2

How to Attempt?

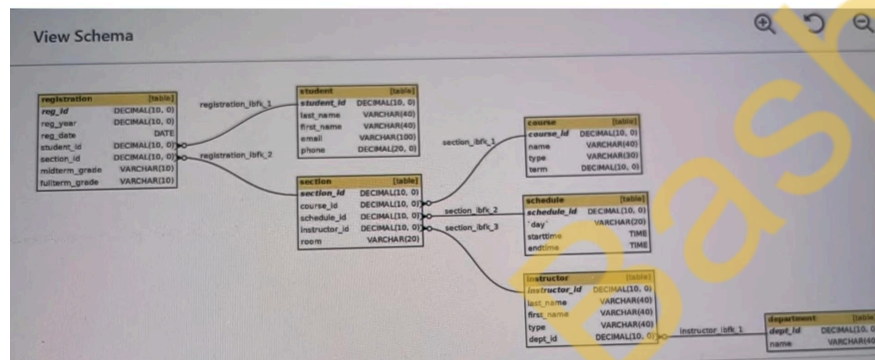
Write an SQL query to display:

The course ID, course name, and schedule details (day and start time) of all courses that are taught on 'Wednesday' (use 'wed' as value from DB table) in rooms. Use Alias "Course ID", "Course Name", "Day", "Start Time"

Your output should have 4 columns as given below:

| Course ID | Course Name | Day | Start Time |
|-----------|-------------|-----|------------|
|-----------|-------------|-----|------------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



Answer :

```
SELECT c.course_id AS "Course ID",
       c.name AS "Course Name",
       s.day AS "Day",
       s.starttime AS "Start Time"
FROM course c
JOIN section sec ON c.course_id = sec.course_id
JOIN schedule s ON sec.schedule_id = s.schedule_id
WHERE s.day = 'wed';
```

Q6.

**Question 2**

**How to Attempt?**

Write an SQL query to display:  
The Title, Price, and ISBN of books that were published after January 1, 1940, and that fall under the "C102" category.

Your output should have three columns as given below:

| Title | Price | ISBN |
|-------|-------|------|
|-------|-------|------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

**Answer :**

```
sql

SELECT Title, Price, ISBN
FROM books
WHERE PublishedDate > '1940-01-01'
AND Category = 'C102';
```

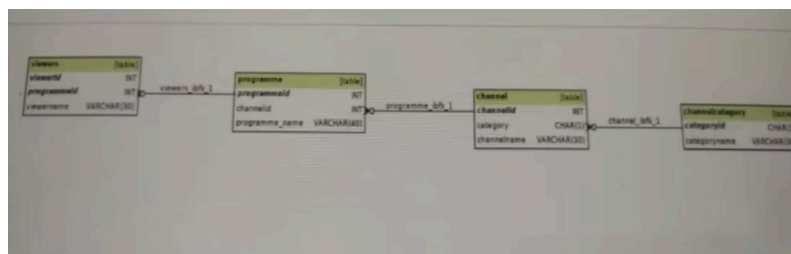
Q7.

Write an SQL query to display:  
The category id and category name whose name starts with 'M'.

Your output should have 2 columns as given below:

| CATEGORYID | CATEGORYNAME |
|------------|--------------|
|------------|--------------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



Q8.

|                           |             |
|---------------------------|-------------|
| <b>train_stations_tbl</b> | [table]     |
| <b>station_id</b>         | VARCHAR(5)  |
| station_name              | VARCHAR(30) |

Q9.

[illegible]

```
SELECT EMPID, LEAVE_TYPE, TOTAL_LEAVES
FROM LEAVE_INFO
WHERE TOTAL_LEAVES > 10
AND LEAVE_TYPE IN ('CL', 'ML');
```

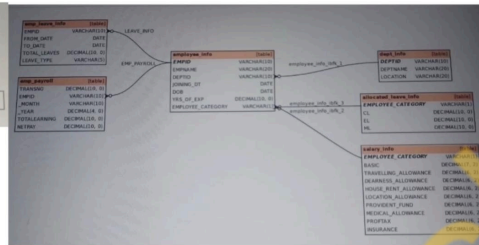
Q10.

Write an SQL query to display:

The ID, Name, Department name, and Basic salary of employees working in the 'HR' department.

Your output should have 4 columns as given below:

| EMPID | EMPNAME | DEPTNAME | BASIC |
|-------|---------|----------|-------|
|-------|---------|----------|-------|



```
SELECT ei.EMPID, ei.EMPNAME, di.DEPTNAME, si.BASIC
FROM employee_info ei
JOIN department_info di ON ei.DEPTID = di.DEPTID
JOIN salary_info si ON ei.EMPLOYEE_CATEGORY = si.EMPLOYEE_CATEGORY
WHERE di.DEPTNAME = 'HR';
```

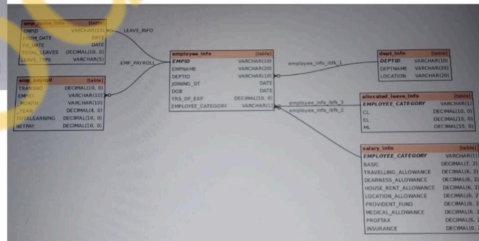
Q11.

Write an SQL query to display:

The employee ID, Name, Department name, and House rent allowance for employees who work in departments located in either BANGALORE or COCHIN.

Your output should have 4 columns as given below:

| EMPID | EMPNAME | DEPTNAME | HOUSE_RENT_ALLOWANCE |
|-------|---------|----------|----------------------|
|-------|---------|----------|----------------------|



```
sql

SELECT ei.EMPID, ei.EMPNAME, di.DEPTNAME, si.HOUSE_RENT_ALLOWANCE
FROM employee_info ei
JOIN department_info di ON ei.DEPTID = di.DEPTID
JOIN salary_info si ON ei.EMPLOYEE_CATEGORY = si.EMPLOYEE_CATEGORY
WHERE di.LOCATION IN ('BANGALORE', 'COCHIN');
```

Q12.

**Question 1**

How to Attempt?

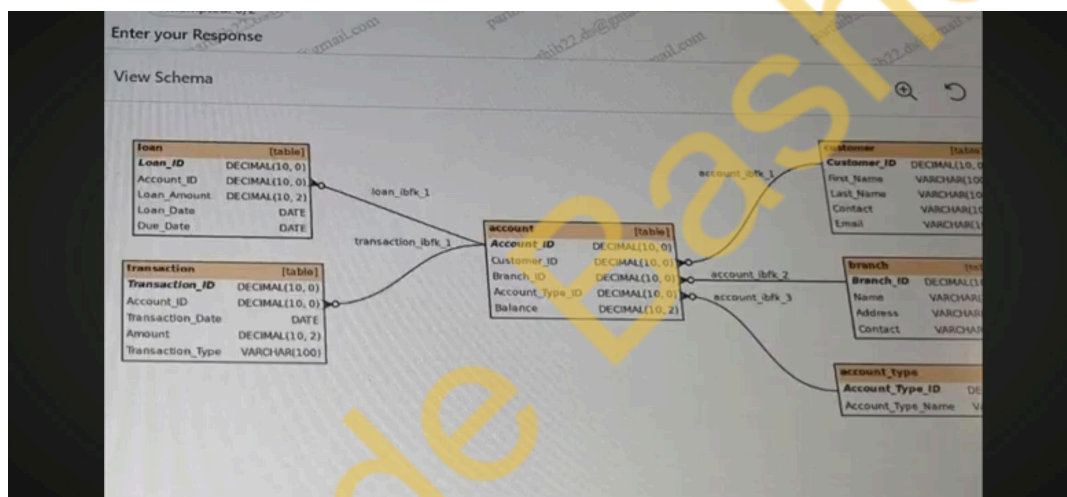
Write an SQL query to display:

The account type id and average account balance for each account type where the average account balance is greater than or equal to 50000.

Your output should have 2 columns as given below:

| Account_Type_ID | Average |
|-----------------|---------|
|-----------------|---------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



```
SELECT Account_Type_ID, AVG(Account_Balance) AS Average
FROM Accounts
GROUP BY Account_Type_ID
HAVING AVG(Account_Balance) >= 50000;
```

Q13.

**Question 2**

How to Attempt?

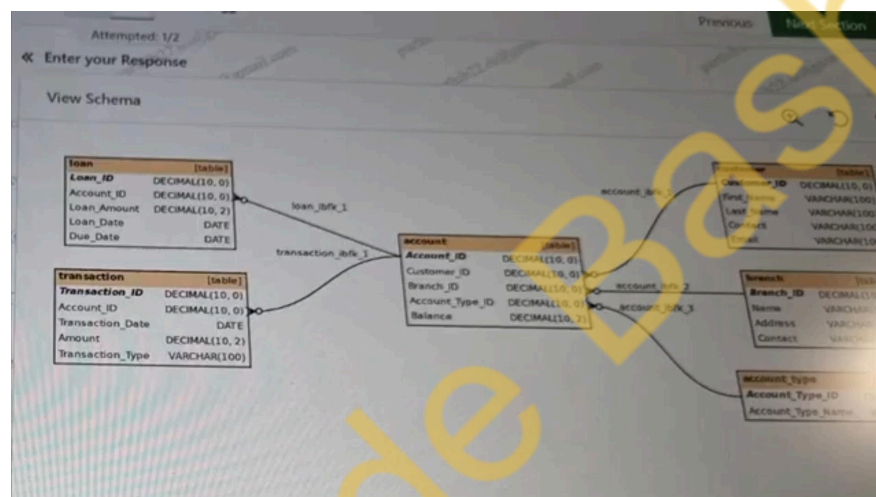
Write an SQL query to display:

First name, last name and account id of customers who have a bank balance greater than or equal to 50000. The output should be ordered by the customer's first name.

Your output should have 3 columns as given below:

| FIRST_NAME | LAST_NAME | ACCOUNT_ID |
|------------|-----------|------------|
|------------|-----------|------------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



Attempted: 2/2

Enter your Response

MYSQL 8.0

```
1 select First_Name as FIRST_NAME,
2 Last_Name as LAST_NAME,
3 Account_ID as ACCOUNT_ID
4 from customer c
5 join account a
6 on c.Customer_ID = a.Customer_ID
7 where a.Balance >= 50000;
```

Q14.

Write an SQL query to display:

The First name (use alias **Staff First Name**), Position and salary of staff members where salary is greater than 50000.

Your output should have three columns as given below:

| STAFF FIRST NAME | POSITION | SALARY |
|------------------|----------|--------|
|------------------|----------|--------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

```
ab.sql
select firstname as "STAFF FIRST NAME", position AS "POSITION", salary AS "SALARY"
from staff
where salary > 50000
```

Q15.

the full name (use alias '**PatientName**'), email (use alias '**PatientEmail**'), admission date (use alias '**AdmissionDate**'), and total billing amount (use alias '**TotalBilling**') for each patient. Include only patients with unpaid bills. Sort the results by the total billing amount in descending order.

Hint : Patientname is combination of first name and last name like "John Doe"

Your output should have 4 columns as given below:

| PATIENTNAME | PATIENTEMAIL | ADMISSIONDATE | TOTALBILLING |
|-------------|--------------|---------------|--------------|
|-------------|--------------|---------------|--------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.

QUESTION 2 SOLUTION

```

1  select
2  |   concat(p.FirstName, ' ', p.LastName) AS
3  PatientName,
4  |   p.Email as PatientEmail,
5  |   p.AdmissionDate AS 'AdmissionDate',
6  |   b.TotalAmount AS TotalBilling
7  FROM
8  |   Patient P
9  JOIN
10 |   Billing b ON p.PatientID =b.PatientID
11 WHERE
12 |   b.PaymentStatus = 'Unpaid'
13 ORDER BY
14 |   b.TotalAmount DESC;

```

Q16.

How to Attempt?

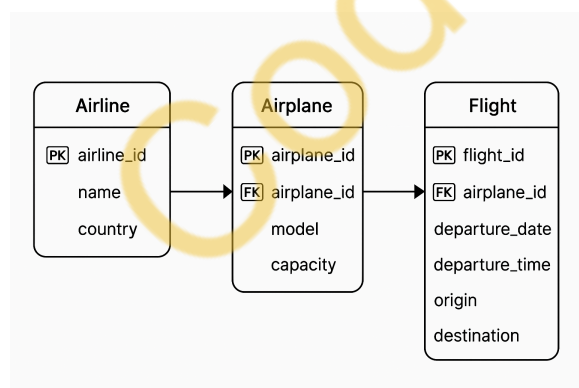
Write an SQL query to display:

List of flights operated by "Singapore Airlines", including the flight ID, departure date and departure time.

Your output should have three columns as given below.

| Flight_ID | Departure_date | Departure_Time |
|-----------|----------------|----------------|
|-----------|----------------|----------------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



```
SELECT
f.FLIGHT_ID,
f.DEPARTURE_DATE,
f.DEPARTURE_TIME
FROM
Flight f
JOIN
Airplane a ON f.AIRPLANE_ID=a.AIRPLANE_ID
JOIN
Airline al ON a.AIRLINE_ID=al.AIRLINE_ID
WHERE
al.NAME='Singapore Airlines';
```

Q17\*.

**Question 1** Revisit Later

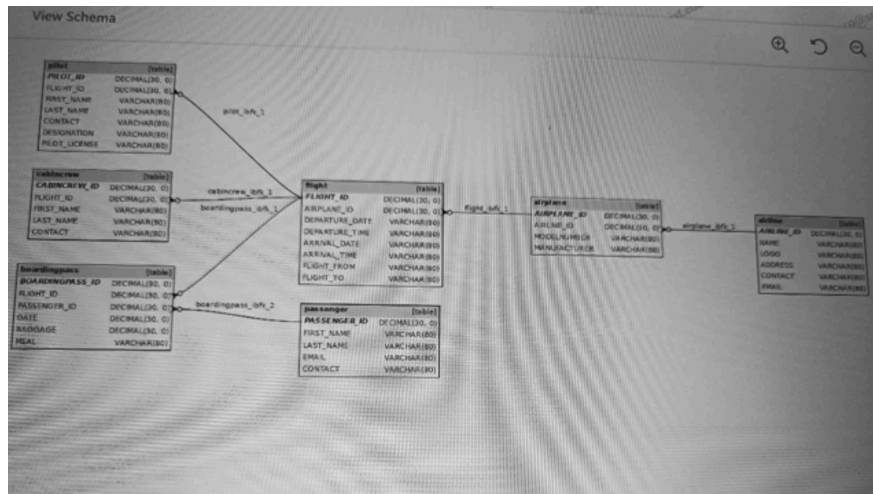
**How to Attempt?**

Write an SQL query to display:  
The airplane ID and model number for airplanes manufactured by Airbus.

Your output should have column as given below:

| AIRPLANE_ID | MODELNUMBER |
|-------------|-------------|
|-------------|-------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right hand side of the screen.



```
SELECT
    AIRPLANE_ID,
    MODELNUMBER
FROM
    Airplane
WHERE
    MANUFACTURER = 'Airbus';
```

Q18\*.

**How to Attempt?**

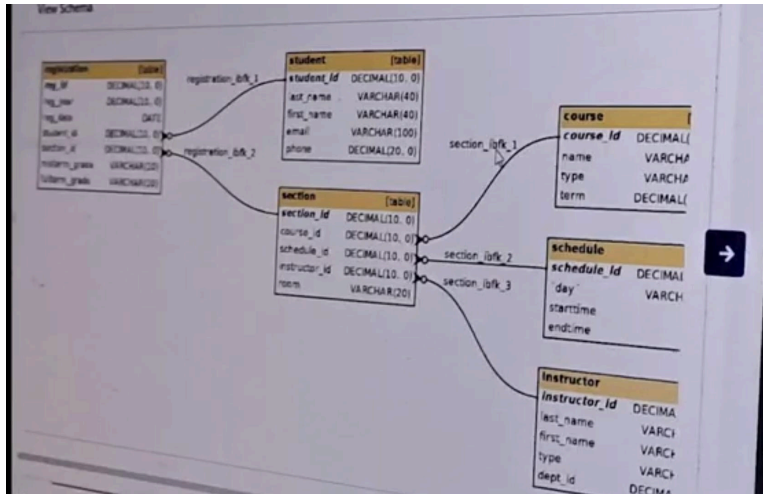
Write an SQL query to display:

The last names of the students registered in the year 2012.

Your output should contain 1 column as mentioned below

last\_name

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.



```

SELECT
    s.LAST_NAME
FROM
    student s
JOIN
    registration r
    ON s.STUDENT_ID = r.STUDENT_ID
WHERE
    EXTRACT(YEAR FROM r.REG_DATE) = 2012;

```

Alternate solution:

```

SELECT
    s.LAST_NAME
FROM
    student s
JOIN
    registration r
    ON s.STUDENT_ID = r.STUDENT_ID
WHERE r.REG_DATE LIKE '2012%'

```

Q19\*.

Question 1

How to Attempt?

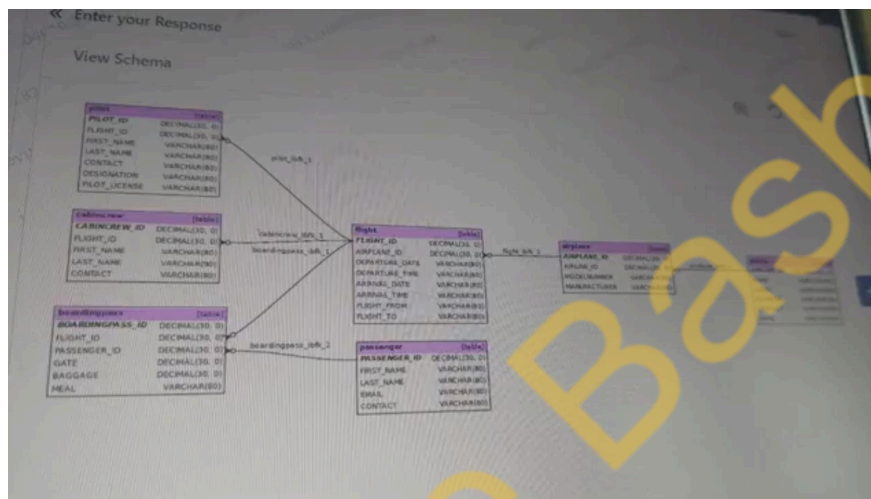
Write an SQL query to display:

The list of cabin crew members whose first names start with the letter 'A' and who are assigned to a flight with a flight number ending with '1'.

Your output should have five columns as given below:

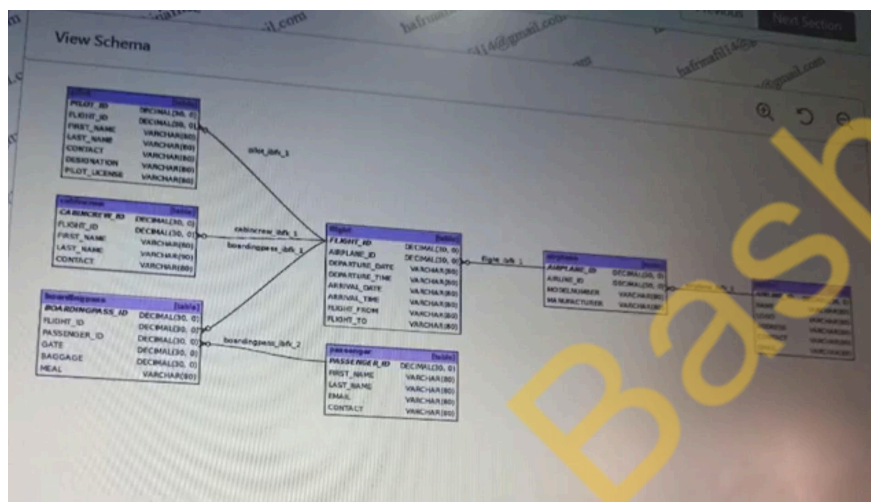
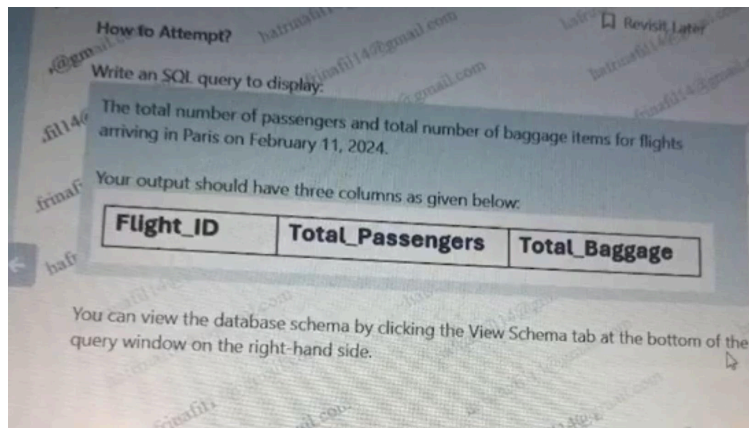
| CabinCrew_ID | First_Name | Last_Name | Contact | Flight_ID |
|--------------|------------|-----------|---------|-----------|
|--------------|------------|-----------|---------|-----------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.



```
SELECT
    c.CABINCREW_ID,
    c.FIRST_NAME,
    c.LAST_NAME,
    c.CONTACT,
    f.FLIGHT_ID
FROM
    cabincrew c
JOIN
    flight f
    ON c.FLIGHT_ID = f.FLIGHT_ID
WHERE
    c.FIRST_NAME LIKE 'A%'
    AND f.FLIGHT_ID LIKE '%1';
```

Q20\*.



```
SELECT
    f.FLIGHT_ID,
    COUNT(bp.PASSENGER_ID) AS Total_Passengers,
    SUM(bp.BAGGAGE) AS Total_Baggage
FROM
    flight f
JOIN
    boardingpass bp
    ON f.FLIGHT_ID = bp.FLIGHT_ID
WHERE
    f.FLIGHT_TO = 'Paris'
    AND f.ARRIVAL_DATE = '2024-02-11'
GROUP BY
    f.FLIGHT_ID;
```

Q21\*.

**How to Attempt?**

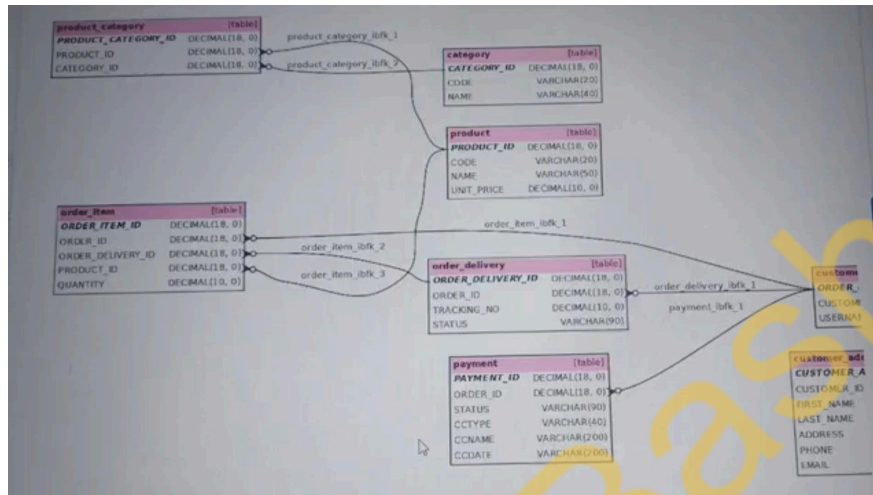
Write an SQL query to display:

The number of women's products available from the table named product (use the alias **product\_count**).

Your output should have 1 column as given below.

| product_count |
|---------------|
|---------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.



```

SELECT
    COUNT(*) AS product_count
FROM
    product p
JOIN
    product_category pc
    ON p.PRODUCT_ID = pc.PRODUCT_ID
JOIN
    category c
    ON pc.CATEGORY_ID = c.CATEGORY_ID
WHERE
    c.NAME = 'Women';
    
```

Q22\*.

**How to Attempt?**

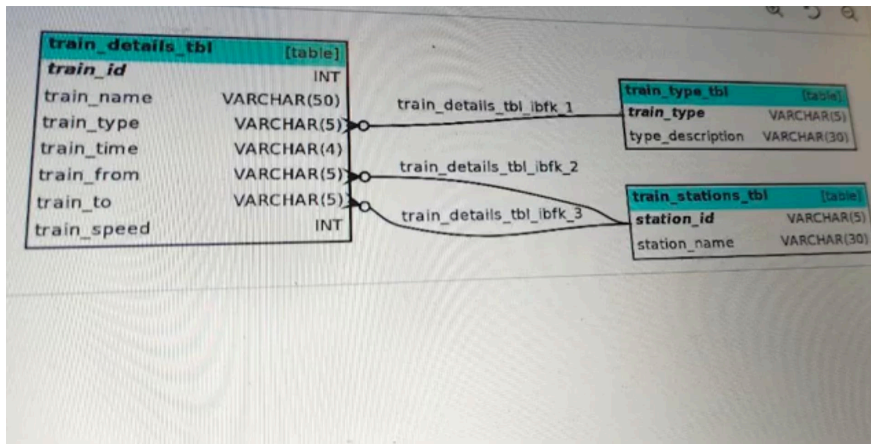
Write an SQL query to display:

The train name and their type which have speed less than 50.

Your output should have 2 column as given below.

| TRAIN_NAME | TRAIN_TYPE |
|------------|------------|
|------------|------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.



```

SELECT
    train_name as TRAIN_NAME,
    train_type as TRAIN_TYPE
FROM
    train_details_tbl
WHERE
    train_speed < 50;
  
```

Q23\*.

How to Attempt?

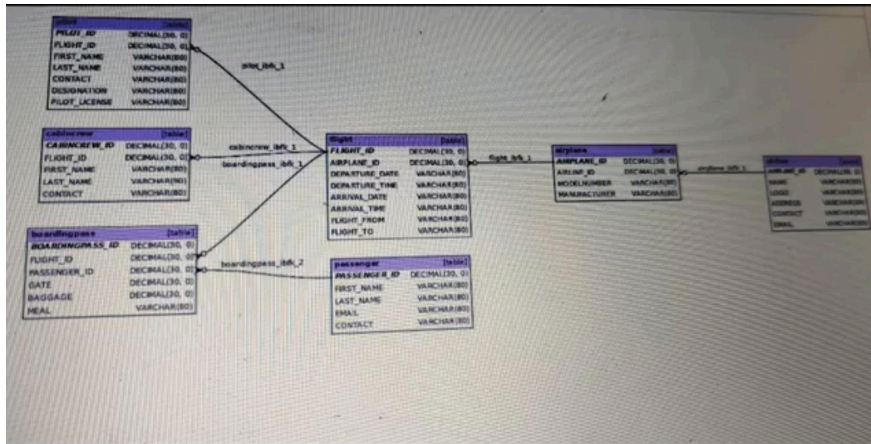
Write an SQL query to display:

The distinct first names and contact numbers of passengers who are departing from 'Hong Kong', boarding a flight with Flight ID '4' and have requested a 'Vegetarian' meal.

Your output should have column as given below.

| FIRST_NAME | CONTACT |
|------------|---------|
|------------|---------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.



```

SELECT DISTINCT
    p.FIRST_NAME,
    p.CONTACT
FROM
    passenger p
JOIN
    boardingpass bp
    ON p.PASSENGER_ID = bp.PASSENGER_ID
JOIN
    flight f
    ON bp.FLIGHT_ID = f.FLIGHT_ID
WHERE
    f.FLIGHT_FROM = 'Hong Kong'
    AND bp.FLIGHT_ID = 4
    AND bp.MEAL = 'Vegetarian';
  
```

Q24\*.

Question 2 Revisit Later

How to Attempt?

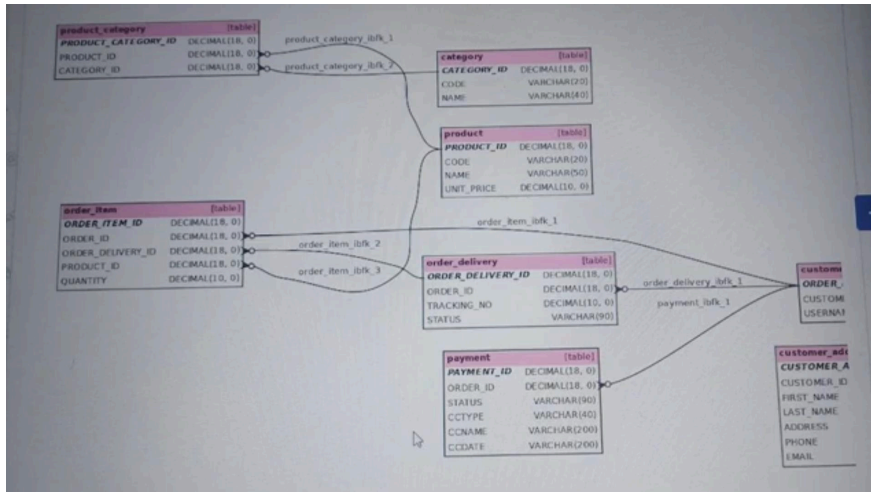
Write an SQL query to display:

The product id and product name of the product(s) whose status are in the transit hub.

Your output should contain 2 columns in the order mentioned below.

| product_id | name |
|------------|------|
|------------|------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.



```

SELECT DISTINCT
    p.PRODUCT_ID,
    p.NAME
FROM
    product p
JOIN
    order_item oi
    ON p.PRODUCT_ID = oi.PRODUCT_ID
JOIN
    order_delivery od
    ON oi.ORDER_DELIVERY_ID = od.ORDER_DELIVERY_ID
WHERE
    od.STATUS = 'In the transit hub';
  
```

Q25\*.

**How to Attempt?**

Write an SQL query to display:

The artist id, name where the artist has a number in the name.

Your output should have 2 columns as given below:

| ARTIST_ID | NAME |
|-----------|------|
|-----------|------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

```
SELECT
    ARTIST_ID,
    NAME
FROM
    artist
WHERE
    NAME REGEXP '[0-9]';
```

Alternate solution

```
SELECT
    ARTIST_ID,
    NAME
FROM
    artist
WHERE
    NAME LIKE '%0%' OR
    NAME LIKE '%1%' OR
    NAME LIKE '%2%' OR
    NAME LIKE '%3%' OR
    NAME LIKE '%4%' OR
    NAME LIKE '%5%' OR
    NAME LIKE '%6%' OR
    NAME LIKE '%7%' OR
    NAME LIKE '%8%' OR
    NAME LIKE '%9%';
```

Q26\*.

Write an SQL query to display:

The message id, content where messages has "Hello" in it.

Your output should have 2 columns as given below:

| MESSAGE_ID | CONTENT |
|------------|---------|
|------------|---------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

```

SELECT
    MESSAGE_ID,
    CONTENT
FROM
    message
WHERE
    CONTENT LIKE '%Hello%';

```

Q27\*.

**How to Attempt?**

Write an SQL query to display:

The full name, license number, and plate number of all drivers whose vehicles have a status of "In Use" and whose number plate ends with "0". The full name should be created by combining the first name and last name with a space between them.

Your output should have 3 columns as given below.

| Name | License_Number | Plate_Number |
|------|----------------|--------------|
|------|----------------|--------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.

```

SELECT
    CONCAT(d.FIRST_NAME, ' ', d.LAST_NAME) AS Name,
    d.LICENSE_NUMBER,
    v.PLATE_NUMBER
FROM
    driver d
JOIN
    vehicle v
    ON d.DRIVER_ID = v.DRIVER_ID
WHERE
    v.STATUS = 'In Use'
    AND v.PLATE_NUMBER LIKE '%0';

```

Q28\*.

Write an SQL query to display:

The license number, vehicle Id, rating, and booking Id of all drivers whose rating is greater than or equal to 4.5 and whose booking status is other than "Cancelled".

Your output should have 4 columns as given below.

| License_Number | Vehicle_ID | Rating | Booking_ID |
|----------------|------------|--------|------------|
|----------------|------------|--------|------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.

```
SELECT
    d.LICENSE_NUMBER,
    v.VEHICLE_ID,
    d.RATING,
    b.BOOKING_ID
FROM
    driver d
JOIN
    vehicle v
    ON d.DRIVER_ID = v.DRIVER_ID
JOIN
    booking b
    ON v.VEHICLE_ID = b.VEHICLE_ID
WHERE
    d.RATING >= 4.5
    AND b.STATUS <> 'Cancelled';
```

Q29\*.

**How to Attempt?**

Write an SQL query to display:

The number of viewers having the same name.

Your output should have 2 columns as given below:

| viewername | name_count |
|------------|------------|
|------------|------------|

You can view the database schema by clicking the **View Schema** tab at the bottom of the query window on the right-hand side of the screen.

```
SELECT
  viewername,
  COUNT(*) AS name_count
FROM
  viewer
GROUP BY
  viewername;
```

Q30\*.

Write an SQL query to display:

Full name (use alias fullname) of the users who have contact with any kind of engineer.

**Note:**

- Full name will be format 'John Doe'
- Any one with the keyword Engineer in their job title is considered as an engineer. (Example - Software Engineer, Mechanical Engineer, Engineer etc.)

Your output should have a column as given below:

| FULLNAME |
|----------|
|----------|

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.

```
SELECT DISTINCT
  CONCAT(u.FIRST_NAME, ' ', u.LAST_NAME) AS FULLNAME
FROM
  users u
JOIN
  contacts c
  ON u.USER_ID = c.USER_ID
JOIN
  users cu
  ON c.CONTACT_ID = cu.USER_ID
JOIN
  jobs j
  ON cu.USER_ID = j.USER_ID
WHERE
  j.JOB_TITLE LIKE '%Engineer%';
```